

SingMaster: A Sight-singing Evaluation System of “Shoot and Sing” Based on Smartphone

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ABSTRACT

Based on the smart phone, this paper integrates OMR (Optical Music Recognition) with sight-singing evaluation, and develops a “shoot and sing” practice APP called SingMaster. This system is mainly composed of three modules: OMR, evaluation and user interface. The OMR module converts the score photographed in the real scene into a note reference sequence. The sight-sing evaluation module first completes the note transcription of the sound spectrum through onset detection and pitch extraction, then aligns the transcribed note sequence with the reference sequence, and performs the evaluation. Finally, the evaluation results are visually fed back to the practitioners through the user interface module. It can provide guidance for practitioners at any time, any place and on any score instead of a real teacher.

CCS CONCEPTS

• Information systems → Music retrieval; • Applied computing → Optical character recognition

KEYWORDS: Optical music recognition, sight-singing transcription, automatic sight-singing system, systematic evaluation measures

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1 INTRODUCTION

With the rapid popularity of smart phones, more and more music intelligent information services are encapsulated in the form of APP. Among these applications, sight-singing APPs are popular among music learners, such as “Sight Singing Studio”. Sight-singing is an ability to read and sing music at first sight [1], which is very important in elementary music education. Our sight-singing practice APP, SingMaster, shows the staff to practitioners and gives intelligent evaluation and feedback like a real teacher, enabling the users to practice anytime and anywhere. Since sight-singing is a compulsory course for professional music students, the sight-singing APPs have a huge user audience.

From the interactive form, the sight-singing APP is divided into “practice mode” and “free style mode”. The practice mode is very similar to karaoke. Users only need to sing the right tone according to the system prompt, without considering the rhythm. In the free style mode, users can control the tone and rhythm of singing as they like. The evaluation in the practice mode only involves the pitch extraction algorithm [2], which has a mature technical scheme at present. The free style mode is a little complex, involving multiple algorithms such as note transcription [3] and alignment [4]. The free style mode is close to the real sight-singing practice environment, just like a teacher face to face. So far, few sight-singing APPs have supported the free style mode like our SingMaster APP.

In addition, the exist sight-singing APPs can only use their own internal score data like music xml, which cannot meet the needs of practicing with textbooks. Our “shoot and sing” system called SingMaster innovatively integrates the OMR [5] and evaluation module. The practitioner takes pictures of the staff with mobile phone, and then sings according to the free style mode. After sight-singing, the system will immediately give an objective evaluation of the pitch and rhythm. This allows practitioners to completely get rid of the dependence on music score content. Whether the score comes from a book, a PDF file or a screenshot, it can be shot and sung immediately, giving users a great deal of freedom.

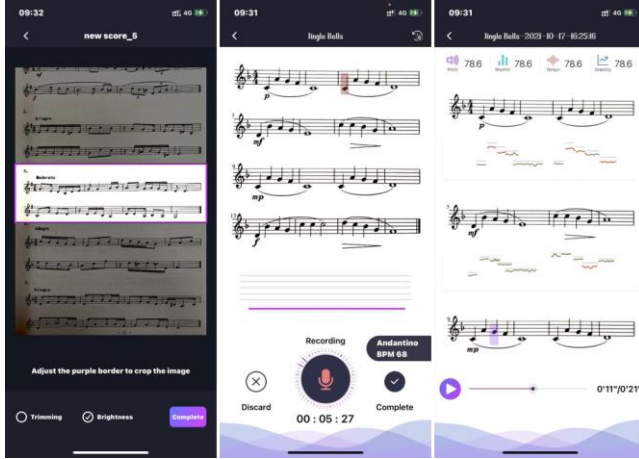


Figure 1: Illustration of our SingMaster APP. Here are three main user interfaces: photographing, singing and feedback.

2 SYSTEM ARCHITECTURE

The SingMaster is mainly composed of three modules: OMR, evaluation and user interface. As shown in Fig. 2, the OMR module converts the score photographed in the real scene into a note reference sequence. The sight-singing evaluation module first completes the note transcription of the sound spectrum through onset detection and pitch extraction, then aligns the transcribed note sequence with the reference sequence, and performs evaluation according to the note information of the reference sequence. Finally, the evaluation results are visually fed back to the practitioners through the user interface module. The feedback contains pitch, rhythm, beat and stability information.

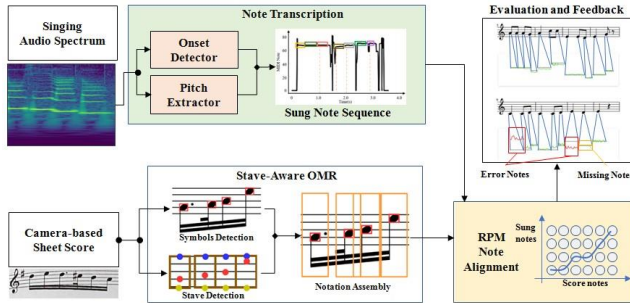


Figure 2: System architecture composed of OMR, evaluation and user interface.

2.1 Stave-aware OMR

The OMR module adopts our own stave-aware OMR method, which adopts an object detection method rather than the CRNN (Convolutional Recurrent Neural Network) sequence recognition [6]. The interference phonemes such as uneven light, angle distortion and spectral plane bending in the photo scene brings great challenges to OMR. Based on the object detection of notes, we introduce the detection of staff, which can effectively improve the recognition accuracy of note pitch and time value. Moreover, this method does not rely on specific training data and has good model generalization ability. On our CPMS (Camera Printed Music

Staves) dataset¹, the pitch accuracy of OMR method perceived by staff is 99.07%, and the time value accuracy is 96.91%, which is better than CRNN sequence recognition method.

2.2 Sight-singing Evaluation

The evaluation module adopts our own multi-stage sight-singing evaluation method [7], which is composed of three steps: onset detection based on CRNN, pitch extraction based on Kim [2], and alignment based on relative pitch. The system aligns the singing note sequence extracted from note transcription with the note sequence extracted from OMR. An objective evaluation score is obtained through the alignment result. We construct the sight-singing vocal dataset² (SSVD). The onset detector achieves 90.61% F-measure. The stages of note transcription and note alignment achieve 88.42% and 94.79% F-measure results, respectively. In addition, the automatic sight-singing system achieves a 77.95% F-measure result on the SSVD dataset.

2.3 User Interface

The user interface module provides feedback of sight-sing evaluation through the graphical user interface. It is divided into two modes: practice mode and evaluation mode. In the practice mode, the system prompts the user with rhythm through the cursor and the vibration sensor of the mobile phone instead of sound. At the same time, the user can see his real-time pitch feedback and quickly adjust his pitch. In the evaluation mode, the system does not have any rhythm and pitch prompt. Detailed visual feedback can be seen in both modes. The correct pitch is a green curve, the wrong pitch is red, and missing singing or insufficient duration is yellow. In addition, the system will also output the speed of sight-singing rhythm and pitch stability information in the form of average value.

3 SYSTEM PERFORMANCE

The SingMaster adopts hybrid deployment. The OMR model is large and deployed on a GPU server using a single NVIDIA GTX1080 card. The evaluation model is small and so can be deployed on the local mobile phone (i.e., Apple iPhone 11). After users take pictures of the music score, our system will automatically upload them to the server. At the same time, users can start to sing and record the audio. These two operations are parallel, so users won't feel the delay of the music score analysis. We test the performance of the above two models. The parsing time of a common score which contains four single staff lines and about fifty notes on our GPU server is 1.39s. The execution time of the evaluation model on the mobile phone is 0.22s. In the future, we will further optimize and tailor the OMR model and deploy the whole system to the mobile phone without relying on the computing support of GPU server.

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¹ <https://github.com/itec-hust/CPMS>

² <https://github.com/itec-hust/Sight-Singing-Vocal-Data>

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